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1. Executive Summary

Kitui Institute of Professionals (KIP) is a growing tertiary college in Kitui County, Kenya, delivering more than 21 examinable programmes across three campuses in Kitui Town, the Kitui CBD and Mwingi Town. As enrolment grows, the college coordination that once fit in a register book and a filing cabinet now strains under its own weight: applications arrive by phone and in person, fee balances are reconciled manually, results move between campuses on paper, and families cannot see progress without travelling to a campus. Each of these frictions slows staff down and weakens the college presentation to prospective students.

This proposal puts forward a complete, integrated college management system for KIP: a public website, online admissions with unique application reference numbers, a student portal, a lecturer workspace, a full administrative console, and an AI study assistant. Crucially, this is not a promise of software to come. The platform has already been built, deployed and is running live on the internet at the address shown below, backed by a managed PostgreSQL database. Every feature described in this document can be inspected and tested today, before any commitment is made.

21+	3	KES 64,000	6–8
Programmes across 3 categories	Campuses, one central system	Total one-off investment	Weeks to full handover

The sections that follow introduce the institution, walk through the platform module by module, provide live demo credentials for each user role, describe the technology and security foundation, and set out the investment: a one-off cost of KES 64,000, recurring hosting of KES 10,000 per year, and maintenance and support at KES 36,000 per year (KES 3,000 per month). A six-phase delivery plan of roughly six to eight weeks takes the college from sign-off to a fully branded, staff-trained go-live.

1.1 What changes for each stakeholder

A platform is only worth owning if it improves the daily experience of the people who use it. The table below summarises the practical change each group of stakeholders will feel in the first term after go-live.

Stakeholder	What changes for them
Prospective students & families	Discover programmes and apply online from any phone; receive an application reference number instantly instead of travelling to a campus to hand in forms
Students	Check fees, results, timetable and learning materials themselves anytime, with an AI study assistant for revision
Lecturers	Marks entered once, saved safely as drafts, published when ready; class rosters always current
Administration & finance	One console for applications, students, fees, payments, results moderation, news and timetables across all three campuses
Management	A single source of truth and an audit trail that shows who did what, and when

Table 1: Stakeholder outcomes after go-live.

2. About Kitui Institute of Professionals

Kitui Institute of Professionals serves eastern Kenya with practical, industry-facing tertiary education. The college prepares students for credentials examined by several national and professional bodies, and it operates from three locations, each serving a distinct community. This footprint is a strength in the market, but it multiplies administrative work: every additional campus adds another place where applications, fee records, timetables and results must be collected, reconciled and shared.

Campus	Location	Setting
Main Campus	100 m from Wikililye, Mulango Road, Kitui	Kitui Town
Town Campus	Muli Mall Building, Kitui CBD	Central business district
Mwingi Campus	Sunrise Building, Mwingi Town	Mwingi Town

Table 2: KIP campuses served by the platform.

The academic offering spans three categories and more than 21 programmes. **Technical** programmes cover ICT, engineering, hairdressing and beauty, and related craft and trade skills. **Business** programmes cover commercial and office practice disciplines. **KASNEB professional** programmes cover the CPA qualification together with CAMS and ATD, examined by KASNEB. Beyond KASNEB, courses are examined by KNEC, NITA and TVET-CDACC, which means records, grading and certification workflows must respect several examining bodies at once. A central digital system is the practical answer for a multi-campus, multi-exam-body college: one source of truth for students, staff and management, accessible from every campus and from home.

2.1 Why now

Today’s students and parents do most of their lives on a phone: money, news, school fees and government services included. A college that can only be reached in person during office hours competes with institutions that answer questions at midnight and accept applications on a Sunday. The gap is not about prestige; it is about expectations.

At the same time, staff time is the college's scarcest resource, and every hour spent re-copying marks or reconciling a fee ledger is an hour not spent teaching or recruiting. Moving these processes online frees that time immediately, and the timing is good: the platform is already built and demonstrable, so the college can act this term rather than budget for a multi-year custom build.

3. The Proposed Solution

The proposed platform is one integrated system rather than a loose collection of tools. Every module shares a single database and a single sign-on, so an application submitted online becomes a student record, which carries fees, results, a timetable and learning materials without any re-typing. Access is role-based: students, lecturers and administrators each see exactly the workspace appropriate to their responsibilities, and nothing else. The table below summarises the six modules and who uses them.

Module	What it delivers	Primary users
Public website	Programme catalogue, campus and contact information, news and announcements, admission calls-to-action	Prospective students, parents, the public
Online admissions	Guided application wizard with validation, unique application reference numbers for every submission, duplicate prevention, next-steps guidance	Applicants, admissions / registrar
Student portal	Personal dashboard, fee statements and balances, published results, timetable, learning materials	Students
Lecturer workspace	Class rosters, marks entry with a draft-to-publish workflow that keeps incomplete marks hidden from students	Lecturers, heads of department
Admin console	Students, applications, courses, fees and payments, results moderation, news and announcements content management, timetable management	Administrators, registrar, finance
AI study assistant	An on-demand study helper inside the portal that answers course questions and guides revision	Students

Table 3: Platform modules at a glance.

Two design decisions deserve emphasis. First, results follow a strict **draft-to-publish** workflow: a lecturer can save work in progress, but students only ever see marks that have been reviewed and published. Second, news and announcements are managed centrally through a small content management system, so one notice posted by the administration appears consistently across all three campuses. Together these keep academic records trustworthy and keep every campus on the same page.

3.1 The student journey, end to end

One journey illustrates how the modules connect. A prospective student finds the college through the public website, browses the programme catalogue and starts the online application. The wizard validates every answer as it is entered, prevents duplicate submissions, and on completion issues a unique application reference number that

the applicant can quote at any campus. When the admissions office accepts the application in the admin console, the applicant becomes a student record in the same database, ready for fees, a timetable and course enrolment. The student then logs into the portal to see a personal dashboard: fee statement and balance, published results as each batch is approved, the class timetable, and learning materials posted by lecturers. Lecturers work in parallel in their own workspace, entering marks as drafts and publishing them once checked, while the administration watches payments, moderates results and posts announcements for all campuses. Nothing is re-typed anywhere in the chain, and every step leaves a record in the audit trail.

4. See It Live: Try the Platform Today

The platform is deployed on Vercel and is running now, with a persistent PostgreSQL database keeping all records safe between visits. You do not have to imagine the system from screenshots: open the address below in any browser, on a phone or a computer, and explore the complete platform.

https://kip-college-system.vercel.app

Live deployment · running on Vercel with a persistent PostgreSQL (Prisma Postgres) database

Three demo accounts are seeded so that you can experience the system exactly as each kind of user would. The accounts are for evaluation; all data behind them is sample data and will be replaced with the college's real records at handover.

Role	Email (username)	Password
Administrator	admin@kipskitui.com	Admin@123
Lecturer	lecturer@kipskitui.com	Lecturer@123
Student	student@kipskitui.com	Student@123

Table 4: Demo credentials for the live deployment.

Suggested things to try

- Log in as the **administrator**: review a submitted application and accept it, post a news announcement for all campuses, moderate results, and inspect fee payments.
- Log in as the **lecturer**: open a class roster, enter marks and save them as a draft, then publish them and watch them become visible to students only after publication.
- Log in as the **student**: check the fee balance and statement, view published results and the timetable, download learning materials, and ask the AI study assistant a question.
- Before logging in, submit a fresh application from the public site and receive your own unique application reference number.

4.1 Notes for the demonstration

The demonstration works on any modern phone, tablet or computer; the interface adapts to the screen, and much of the college's audience will first meet the system on a phone. If you submit a test application as suggested above, log in afterwards as the administrator to see it arrive in the applications queue, then accept it and watch it become a student record. When the demonstration is complete, all test data can be cleared so that go-live starts from a clean, branded system loaded only with the college's real programmes and staff accounts.

5. Technology & Security

The system is built on a modern, well-supported technology stack, chosen so that any competent developer in Kenya can maintain it for years to come. The application is deployed on Vercel, a global hosting platform, with the PostgreSQL database automatically provisioned and managed alongside it; the college needs no server room, no on-site hardware and no dedicated IT staff to keep it online.

Layer	Technology
Application framework	Next.js 16 (React, server-side rendering)
Language	TypeScript, strict mode
Interface	Tailwind CSS with shadcn/ui components
Database	PostgreSQL via Prisma Postgres (auto-provisioned, persistent)
Hosting & deployment	Vercel, with continuous deployment from a version-controlled repository

Table 5: Technology stack summary.

Security was designed in from the start rather than bolted on. Every route in the system is protected by a combination of the measures below, and the platform ships with an audit trail so that administrators can review who did what and when.

- **Role-based access control**: admin, lecturer and student boundaries are enforced on the server, not just hidden in the interface.
- **Rate limiting**: login attempts, password changes, application submissions and assistant queries are throttled to blunt brute-force attacks and abuse.
- **Audit trail**: security-relevant events, including logins, password changes, application decisions and results publication, are recorded for later review.
- **Secure sessions**: hardened session cookies, with expired sessions cleaned up automatically.
- **Strict input validation** on every form and API route, plus protective security headers on all responses.

5.1 Reliability, data ownership and backups

Vercel runs the application on a global network with automatic scaling and HTTPS encryption everywhere, so pages stay fast even on modest mobile connections, and the managed PostgreSQL database persists every record between visits. The college owns its data outright: records can be exported at any time, and the application itself

lives in a version-controlled repository whose full history shows every change ever made. The database is professionally managed and backed up as part of the hosting service, which means a lost laptop, a stolen office computer or a power cut in Kitui Town cannot take the college's records with it. Nothing about the system ties the college to a single developer: the stack is mainstream, the code is documented, and any competent developer can take it over.

6. Investment & Payment Schedule

The total one-off investment to own this platform outright is **KES 64,000**, covering design and branding, development, content, testing and deployment, and staff training. There are no licence fees and no per-student charges, and the college keeps full control of its code and data. The breakdown is shown below.

Item	What it covers	Amount (KES)
Design & branding	Visual identity for the platform, colour system, typography, layout polish	15,000
Platform development	Build, configuration and integration of all six modules on the live system	22,000
Content, testing & deployment	Real programme content, end-to-end testing, production deployment on Vercel	18,000
Training & handover	Staff training sessions, documentation, admin credentials and knowledge transfer	9,000
Total one-off investment		**64,000**

Table 6: One-off investment breakdown (total KES 64,000).

Payment is staged against visible progress rather than paid up front. Half is due when the proposal is signed off, a quarter when the staging site with the college's own branding and content is delivered for review, and the final quarter at go-live, when the platform is fully operational and staff are trained.

Milestone	Trigger	Share	Amount (KES)
First payment	Proposal sign-off	50%	32,000
Second payment	Staging delivery (branded, content-loaded review site)	25%	16,000
Final payment	Go-live (fully operational, staff trained)	25%	16,000
Total		**100%**	**64,000**

Table 7: Payment milestones.

7. Recurring Costs: Hosting & Support

Beyond the one-off investment, two recurring services keep the platform online and healthy. Hosting covers the Vercel deployment, the managed PostgreSQL database, the custom domain and SSL encryption. Maintenance and support covers updates, dependency and security patches, bug fixes, small content changes and a support channel for staff, billed at KES 3,000 per month.

Service	Frequency	Cost (KES)	What it includes
Hosting	Per year	10,000	Vercel hosting, managed PostgreSQL database, domain, SSL
Maintenance & support	Per year (KES 3,000 per month)	36,000	Updates and patches, bug fixes, content tweaks, staff support channel
Total recurring	**Per year**		**46,000**

Table 8: Recurring costs after go-live.

Both recurring items are payable annually and can be reviewed together each year as the college's needs grow. If the college prefers to start leanly, hosting alone keeps the platform public and available; the support retainer can be added at any point in the year.

8. Delivery Timeline

Because the platform already exists and is running, delivery is not a software construction project; it is a focused programme of branding, content, verification and training. The plan below spans approximately six to eight weeks from sign-off to a fully operational, staff-trained go-live, with the second and third payment milestones tied to staging delivery and go-live respectively.

Phase	Focus	Timing
1. Discovery & sign-off	Confirm requirements, programme list, branding assets and fee figures; sign agreement; first milestone payment	Week 1
2. Branding & design	Apply KIP identity across the site and portals; refine layouts and imagery	Weeks 1–2
3. Configuration & data setup	Load real programmes, campuses, courses, fee structures and staff accounts	Weeks 2–4
4. Content, testing & staging	Populate content, end-to-end testing, staging delivery for college review; second milestone payment	Weeks 5–6
5. Training & handover	Train administrators, lecturers and front-desk staff; hand over credentials and documentation	Week 7
6. Go-live & support transition	Cut over to production domain, monitor, transition to the support retainer; final milestone payment	Week 8

Table 9: Six-phase delivery plan (approximately 6–8 weeks).

8.1 What the college supplies

The plan above assumes a small number of inputs from the college, all of which are one-time tasks rather than ongoing work. In Phase 1 the college confirms the authoritative programme list with fee figures and the correct contact details, and supplies branding assets such as the logo and preferred colours. In Phase 3 the administration nominates the staff who need accounts, together with their roles, and agrees the fee structures to load for each programme. Everything else, from configuration and testing to deployment and training, is handled as part of the delivery. In practice a responsive college completes its inputs within a few working days of each request, which is why the eight-week end-to-end plan carries comfortable slack.

9. Next Steps & Contacts

The fastest route to value is a short decision loop. We recommend reviewing the live platform with two or three senior staff, confirming the facts listed for verification, then signing off so that branding and data work can begin immediately. The college can be running its own system before the next intake.

1. Explore the live platform at **https://kip-college-system.vercel.app** using the demo accounts in Section 4, ideally with the registrar, finance and a senior lecturer present.
2. Confirm the facts pending owner verification: contact telephone numbers, fee figures, and the authoritative programme list.
3. Sign off this proposal and make the first milestone payment of KES 32,000 so that Phase 1 can begin.
4. Supply branding assets (logo, colours) and nominate staff for training in Phase 5.
5. Approve the staging site at the second milestone and schedule go-live.

Contact for this proposal

Email: **info@kipskitui.com**

Phone: **+254 703 710 879** (to be verified by owner)

** Note: telephone numbers and fee figures quoted in this proposal are pending verification by the college owner and should be confirmed before the document is circulated externally or used in marketing.*